

FU Xiao

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EDUCATION

Zhejiang University (ZJU)

Hangzhou, China

Bachelor of Information Eng. | College of Information Science & Electronic Engineering (ISEE) Sept. 2018 – Jul. 2022

- **GPA:** 3.96/4.00, 90.63/100, **Ranking:** 1/144
- **Laureate of National Scholarship (twice), Graduated with Dean's Honor**
- **Honors Program:** ZJU Morningside Cultural China Scholars Program (36 students shortlisted from 12,000 freshmen with distinguished innovative capability and leadership)
- **Research Interest:** AI, Big Data and Multimedia Processing, especially 3D Computer Vision

PUBLICATIONS

* equal contribution, † corresponding author

- [Xiao Fu](#)*, Shangzhan Zhang*, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao†
“**Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation**”
In *Proceedings of the International Conference on 3D Vision (3DV)*, 2022, Prague, Czechia, pp. 301-311, doi: 10.1109/3DV57658.2022.00042 [[Project Page](#) | [Paper](#) | [Poster](#) | [Video](#) | [Code](#)]
- Guangyi Zhang*, [Xiao Fu](#)*, Qiyu Hu†, Yunlong Cai, Guanding Yu
“**Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network**”
In *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2021, Helsinki, Finland, pp. 678-683, doi: 10.1109/PIMRC50174.2021.9569633 [[Paper](#) | [Slides](#)]
- Tong Wu, Jiarui Zhang, [Xiao Fu](#), Yuxin Wang, Jiawei Ren, Liang Pan, Wayne Wu, Lei Yang, Jiaqi Wang, Chen Qian, Dahua Lin, Ziwei Liu
“**OmniObject3D: Large-Vocabulary 3D Object Dataset for Realistic Perception, Reconstruction and Generation**” [[Paper](#)]
Under Review of *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023 (ID : 2642)

HONORS & AWARDS

- National Scholarship, Ministry of Education of P.R. China (**Top 0.2% across China, twice**) 2020, 2021
- ISEE Excellent Student Award (**Top 5 students among ISEE 314 undergraduates**) 2021
- ISEE Yuelun Alumni Scholarship (**Dean's Award for Academic Contest, Bonus 20,000 CNY**) 2020
- First-Class Scholarship for Academic Excellence of Zhejiang University (**Top 3%**) 2020, 2021
- Government Scholarship for Academic Excellence of Zhejiang Province (**Top 3%**) 2019
- Outstanding Graduation Thesis Award of Zhejiang University 2022
- Finalist Winner, Mathematical Contest in Modeling (MCM/ICM) (**Top 2% out of 20,954 teams**) 2020
- Meritorious Winner, Mathematical Contest in Modeling (MCM/ICM) (**Top 7% out of 26,112 teams**) 2021
- Gold Prize, 7th “Internet Plus” College Student Innovation and Entrepreneurship Contest (1st author) 2021
- Gold Prize, 12th “Challenge Cup” National College Student Business Plan Competition, National 2020
- Grand Prize, 12th “Challenge Cup” National College Student Business Plan Competition, Provincial 2020
- Grand Prize, “Yongdian Cup” Innovation and Entrepreneurship Contest (1st author) 2019
- Second Prize, College Student Physical Innovation Competition (Theoretical) of Zhejiang Province 2020
- Third Prize, 17th “Challenge Cup” Extracurricular Academic Science and Technology Contest (1st author) 2021
- Outstanding Student Award of Zhejiang University 2019, 2020, 2021
- Academic Excellence Scholarship of Zhejiang University 2019, 2020, 2021

RESEARCH EXPERIENCE

Neural Panoptic 3D Reconstruction of Large-Scale Real-World Scenario

Shanghai AI Laboratory

Mentor: [Prof. Dahua Lin](#) and [Dr. Jiaqi Wang](#)

Aug. 2022 – Now

- generated high-quality scene reconstruction and panoptic labels in 3D space from multi-view images and semantic label results predicted by 2D pre-trained network on the ScanNet dataset
- leveraged the coarse semantic segmentations of 2D images and the SDFs of scene reconstruction to obtain the accurate instance labels in 3D space, especially in overlapped regions of objects
- employed the reconstruction priors to improve the accuracy of 3D semantic/instance segmentation masks
- intending to publish a pertinent paper to ICCV'23 in capacity of the first author

OmniObject3D: A Large-Vocabulary and Realistic 3D Object Dataset

Shanghai AI Laboratory

Mentor: *Prof. Dahua Lin and Dr. Tong Wu*

Jul. 2022 – Nov. 2022

- proposed OmniObject3D, a large-vocabulary 3D object dataset with massive high-quality real-scanned 3D objects
- (1) Large Vocabulary: comprises 6,000 scanned objects in 190 daily categories, sharing common classes with popular 2D datasets, benefiting the pursuit of generalizable 3D representations
- (2) Rich Annotations: each 3D object is captured with both 2D and 3D sensors, providing textured meshes, point clouds, multi-view rendered images, and multiple real-captured videos
- (3) Realistic Scans: provides professionally scanned object scans with precise shapes and realistic appearances
- set up four tracks: robust 3D perception, novel-view synthesis, neural surface reconstruction, and 3D generation

Unaligned Sparse Object-Centric Image Synthesis

Shanghai AI Laboratory

Mentor: *Dr. Jiaqi Wang and Dr. Sida Peng*

Mar. 2022 – Jul. 2022

- focused on view synthesis from sparse inputs on unaligned coordinate system, a more generic scenario, where category-level priors can not be stored by *xyz* inputs
- projected the features of source views in 3D space via monocular depth predicted by MiDaS and replaced *xyz*, from which network can implicitly learn priors, in NeRFormer with nearest features at novel viewpoints
- outperformed NeRFormer w/o *xyz* and ViewFormer in terms of novel view synthesis, and was comparable with NeRFormer in case of sparse views (1-3 views)

Panoptic NeRF: 3D-to-2D Panoptic Label Transfer

Max Planck Institute for Intelligent Systems & ZJU

Mentor: *Prof. Andreas Geiger and Prof. Yiyi Liao*

Sept. 2021 – Mar. 2022

- proposed to utilize neural rendering as a tool to generate high-quality 2D panoptic segmentation annotations from a set of sparse images and easy-to-obtain coarse 3D bounding primitives and noisy 2D semantic predictions
- leveraged a novel dual formulation of the semantic fields to improve the underlying geometry reconstruction and resolve label ambiguity at the intersection region of the 3D bounding primitives
- reduced the label annotation time from 1.5 hours to 0.75 minute per frame in a single street scene
- achieved SOTA label transfer results in terms of accuracy and multi-view consistency on urban scenes of the KITTI-360 dataset and enabled rendering of multi-view and spatio-temporally consistent labels at novel viewpoints

Algorithm-Based Dual-Layer Deep-Unfolding Neural Network

ZJU

Mentor: *Prof. Guanding Yu*

Jun. 2020 – Feb. 2021

- proposed a learning-based framework to address the high computational complexity problem of a convex optimization algorithm, the dual-layer iterative Penalty Dual Decomposition (PDD) algorithm
- unfolded the PDD algorithm into a layer-wise structure whereby some trainable parameters were introduced into the places of the high-complexity operations, e.g., large-dimensional matrix inversions and iterative sub algorithms
- presented high performance of the PDD algorithm ($\geq 97\%$) with a greatly reduced complexity, i.e., total number of iterations ($10,000 \rightarrow 70$), optimization of $\mathbf{V}_{RF}$ ($\mathcal{O}(N^2 N_{RF}^2) \rightarrow \mathcal{O}(1)$) and $\mathbf{U}_{RF}$ ($\mathcal{O}(KM^2 M_{RF}^2) \rightarrow \mathcal{O}(1)$), pseudo matrix inversion $\rightarrow$ matrix multiplication ($\mathcal{O}(N^3) \rightarrow \mathcal{O}(N^{2.37})$)
- focused on solving the spectrum efficiency maximization problem for hybrid pre-coding architecture
- **Laureate of Excellent Award (Team leader) - Student Research Training Program (Province-level)**

EXTRACURRICULAR ACTIVITIES

ZJU Volunteer Organization

Hangzhou, China

Participant

Sept. 2018 – Jun. 2022

- contributed with over 250 volunteer hours by participating in more than 30 volunteer services, such as the “Chunlin” Anti-Epidemic Online Teaching activity, Prince Bay guidance service, “Internet Plus” reception service, ZJU autumn sporting events, and hometown anti-epidemic activities
- **Laureate of ZJU Five-Star Volunteer, the highest honor awarded for ZJU volunteer participants**

ISEE Youth League Committee

Hangzhou, China

Deputy Secretary

Sept. 2020 – Jul. 2021

- assisted the league committee teachers in carrying out academic competitions as well as innovation and entrepreneurship competitions (“Internet Plus” and “Challenge Cup”)
- participated in the sharing session of outstanding seniors of ISEE, and the counterparts of the 19 grade assembly, epidemic prevention, and employment quality analysis
- **Laureate of Excellent Member Award of ZJU Youth League Committee**

EXTRACURRICULAR SKILLS

Computer: Python, PyTorch, TensorFlow, C/C++, MATLAB, Java, LaTeX, Linux, Caffe.

Language: TOEFL (Internet-based) : 99 (R:27,L:25,S:22,W:25)